Clinical Outcome Prediction from Noisy Medical Records

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Abstract

Hospital readmission prediction is an important healthcare problem because early identification of high-risk patients helps hospitals improve patient care, reduce avoidable readmissions, and support better clinical decision-making. In this project, we use the **Diabetes 130-US Hospitals** dataset, which contains real clinical records collected from 130 hospitals in the United States between 1999 and 2008. The dataset includes 101,766 records and 50 features related to demographics, diagnoses, medications, laboratory results, and hospital visits. The goal is to predict whether a diabetic patient will be readmitted within 30 days. The project addresses realistic medical data challenges including missing values, noisy records, duplicate patient encounters, and severe class imbalance. Preprocessing included missing value handling, duplicate removal, categorical encoding, feature selection via Random Forest importance, StandardScaler, and PCA reduction to 23 components. To simulate real-world label errors, 10% label noise was injected as an adversarial condition. Seven baseline models were trained and compared: Logistic Regression, Decision Tree, Random Forest, Naive Bayes, KNN, SVM, and Gradient Boosting. Models were evaluated using Accuracy, Recall, F1-score, and AUC-ROC. **Logistic Regression achieved the best overall clinical performance** with AUC = 0.622, Recall = 0.53, and F1 = 0.21 — outperforming models that showed deceptively high Accuracy but failed entirely to detect the at-risk minority class. The results confirm that proper evaluation metrics are essential when working with imbalanced healthcare datasets.

1 Introduction and Motivation

Hospital readmission is one of the most important quality indicators in healthcare systems. Patients who are readmitted within a short period after discharge often indicate incomplete treatment, poor follow-up care, or complications in disease management. For diabetic patients, hospital readmission is especially common due to the long-term nature of the disease and the presence of multiple related health conditions.

Predicting hospital readmission helps hospitals identify high-risk patients earlier and allows medical teams to improve discharge planning, follow-up treatment, and resource allocation, reducing hospital costs and improving patient outcomes.

In this project, we focus on predicting **30-day hospital readmission** using the **Diabetes 130-US Hospitals dataset**. Unlike clean benchmark datasets, this dataset contains realistic medical challenges such as missing values, duplicate records, noisy labels, and severe class imbalance, making it suitable for building a practical machine learning solution.

Because the dataset is severely imbalanced, Accuracy alone is a misleading metric — a model that always predicts 'not readmitted' achieves over 90% Accuracy while completely failing the clinical task. We therefore prioritize **Recall, F1-score, and AUC-ROC** as primary evaluation criteria, since detecting the at-risk minority class is the core clinical objective.

We also introduce an adversarial condition by injecting 10% label noise to test model robustness under realistic hospital data errors such as incorrect diagnoses or recording mistakes.

2 Related Work

Many previous studies have applied machine learning techniques to hospital readmission prediction and healthcare classification tasks. Commonly used models include Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, and Gradient Boosting methods.

Logistic Regression is frequently cited as a strong baseline for medical prediction tasks due to its interpretability and stable performance in imbalanced settings. Medical staff need to understand why a prediction was made, making interpretable models especially valuable in clinical deployment. Logistic Regression also handles class imbalance effectively when combined with balanced class weights.

Random Forest and Gradient Boosting are often preferred for higher predictive power, as they capture nonlinear relationships. However, on severely imbalanced datasets these models can achieve deceptively high Accuracy by predicting only the majority class, while producing near-zero Recall for the minority class — a critical failure in medical applications.

Previous work strongly emphasizes that healthcare evaluation must rely on Recall, F1-score, and AUC-ROC rather than Accuracy alone. A model with 91% Accuracy but 0% Recall for the readmitted class is clinically useless, as it misses all high-risk patients. Our project follows these best practices and extends them by adding adversarial label noise to evaluate model robustness under realistic hospital data quality problems.

3 Data Description and Preprocessing

3.1 Dataset

The dataset used is the **Diabetes 130-US Hospitals** dataset. It contains 101,766 patient encounters collected from 130 US hospitals between 1999 and 2008, with 50 features covering demographics, diagnosis codes, medication usage, laboratory results, and clinical history. The dataset is available at:
https://www.kaggle.com/code/chongchong33/predicting-hospital-readmission-of-diabetics

The target variable *readmitted* originally contains three classes: **NO**, **>30**, and **<30**. It was converted into **binary classification**:

- 0 = Not readmitted within 30 days (majority class)
- 1 = Readmitted within 30 days (<30) — the clinically critical positive class (minority)

3.2 Preprocessing Pipeline

The following preprocessing steps were applied sequentially:

- Missing values represented by '?' were replaced with NaN.
- The *weight* column was removed due to ~97% missing values.
- Columns *race*, *payer_code*, and *medical_specialty* were filled with 'Unknown'; *max_glu_serum* and *A1Cresult* filled with 'None' to preserve the clinical meaning that the test was not performed.
- Duplicate patient encounters reduced by keeping only the first visit per patient, preventing data leakage.
- Identifier columns and near-zero variance medication columns were removed.
- Categorical variables were encoded using ordinal, binary, and one-hot encoding.
- StandardScaler was applied for feature normalization.
- Random Forest feature importance was used to select the most predictive features.
- PCA reduced the dataset to **23 principal components**, preserving ~90% of variance.

4 Methodology

4.1 Data Splitting and Adversarial Condition

The dataset was split into training, validation, and test sets using a **70% / 15% / 15%** ratio. The validation set was used for model comparison and hyperparameter tuning; the test set was reserved only for final evaluation.

An **adversarial label noise** condition was introduced by randomly flipping 10% of the target labels (6,997 labels), simulating incorrect diagnoses or data entry mistakes in hospital systems, and testing model robustness under realistic data quality problems.

4.2 Baseline Models

Seven baseline classifiers were trained and compared:

- Logistic Regression
- Decision Tree
- Random Forest
- Naive Bayes
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Gradient Boosting

4.3 Evaluation Metrics

Because the dataset is heavily imbalanced, **Recall, F1-score, and AUC-ROC were treated as the primary evaluation metrics**. Accuracy was recorded for completeness but not used for model selection. Confusion matrices and ROC curves were also used to analyse minority-class detection performance.

4.4 Hyperparameter Tuning

The three best-performing models — Logistic Regression, Random Forest, and SVM — were selected for hyperparameter tuning using GridSearchCV with *class_weight='balanced'*. The best Logistic Regression configuration was C = 0.01, solver = lbfgs.

5 Experimental Results and Discussion

5.1 The Accuracy Trap in Imbalanced Data

The baseline comparison revealed a critical failure mode: several models achieved very high Accuracy while completely failing the clinical task. **Gradient Boosting reached 91.0% Accuracy yet produced Recall = 0.00 and F1 = 0.00** for the readmitted class, meaning it never identified a single at-risk patient. KNN showed a similar pattern (Accuracy = 90.2%, Recall = 0.02, F1 = 0.04). These models learned to always predict the majority class, a classic imbalanced-data failure. In a medical setting this is clinically dangerous and these models were excluded from final selection.

5.2 Full Model Comparison

Table 1 summarises all seven baseline models across the four primary metrics:

Model	Accuracy	AUC	Recall (Readmitted)	F1 (Readmitted)
Logistic Regression	**0.636**	**0.621**	**0.53** ✓	**0.21** ✓
Decision Tree	0.490	0.592	0.50	0.18

Random Forest	0.742	0.610	0.46	0.20
SVM	0.693	0.619	0.44	0.20
Naive Bayes	0.865	0.594	0.08	0.11
KNN	0.905	0.535	0.02 ✗	0.04 ✗
Gradient Boosting	0.910	0.617	0.00 ✗	0.00 ✗

Table 1: Baseline Model Comparison. ✓ Best minority-class performance. ✗ Model fails to detect readmitted patients despite high Accuracy.

5.3 Best Model: Logistic Regression

Logistic Regression achieved the best overall clinical performance across all meaningful evaluation metrics. With AUC = 0.622, Recall = 0.53, and F1 = 0.21, it is the only model that reliably detects the at-risk minority class. Table 2 shows its final test-set results after hyperparameter tuning:

Metric	Value
Test Accuracy	0.631
Test AUC	0.622
Recall — Readmitted Class	0.53
F1-score — Readmitted Class	0.21
Best Hyperparameters	C = 0.01, solver = lbfgs

Table 2: Logistic Regression — Final Test Results (Best Model)

Logistic Regression was selected as the final best model for the following reasons:

- **Best Recall (0.53)** — detects 53% of at-risk patients; competing high-Accuracy models detect near zero.
- **Best F1-score (0.21)** — the only model with a meaningful minority-class F1.
- **Strong AUC (0.622)** — competitive discrimination ability overall.
- **Clinical interpretability** — feature coefficients can be understood and explained to medical staff.
- **Handles class imbalance** — class_weight='balanced' allows learning from the minority class.

5.4 Hyperparameter Tuning

GridSearchCV was applied to Logistic Regression, Random Forest, and SVM. The best Logistic Regression configuration (C = 0.01, solver = lbfgs) produced only marginal AUC improvement over the baseline, indicating that strong preprocessing — particularly PCA and feature selection — was already providing most of the benefit. This highlights that data quality and preprocessing can be more impactful than extensive hyperparameter search.

5.5 Adversarial Label Noise

After flipping 10% of training labels, all models showed a consistent performance drop. This demonstrates how sensitive healthcare prediction models are to data quality issues such as incorrect diagnoses or recording mistakes, underscoring the need for data validation before clinical deployment.

5.6 Feature Analysis

Feature importance analysis revealed clinically meaningful patterns. The variable *number_inpatient* — number of previous inpatient visits — was the strongest predictor of readmission. Age was also a strong predictor, particularly

for patients aged 60–90. These findings are consistent with clinical literature and support the practical validity of the model.

6 Conclusion and Future Work

This project successfully applied supervised machine learning to predict 30-day hospital readmission using noisy clinical records from diabetic patients. The study demonstrated that proper preprocessing, balanced evaluation metrics, and robustness testing are critical when working with real-world healthcare data.

Logistic Regression was selected as the final best model because it achieved the highest clinically meaningful performance: AUC = 0.622, Recall = 0.53, and F1 = 0.21. Although Gradient Boosting achieved the highest raw Accuracy (0.910), it produced Recall = 0.00 and F1 = 0.00 for the readmitted class, rendering it clinically useless. This confirms that **Accuracy alone is a misleading metric for imbalanced healthcare datasets**.

The adversarial label noise experiment confirmed that noisy labels reduce model reliability across all models, emphasising the importance of data quality in medical prediction systems. Feature analysis identified *number_inpatient* and patient age as the strongest predictors of early readmission.

Future work could improve the system through:

- SMOTE or other oversampling techniques to further address class imbalance.
- Threshold tuning to optimise the Recall/Precision trade-off for clinical deployment.
- Voting ensembles and stacking to combine model strengths.
- Explainable AI methods such as SHAP values for clinical interpretability.
- Integration of additional clinical features and external validation datasets.

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